

# Conceptual Modelling of Legal Compliance Requirements - The cases of eIDAS2 and NIS2

PAULO BOLINHAS, Instituto Superior Técnico, PT

## ABSTRACT

This work explores the hypothesis that conceptual models can simplify the understanding of complex legal compliance requirements. There is a recognized absence of complete and well-defined models for eIDAS2 (Electronic Identification, Authentication, and Trust Services) and NIS2 (Network and Information Systems), two critical pieces of European legislation addressing digital identity and cybersecurity, respectively. Through conceptual modelling, this research fills both gaps by developing two detailed architecture descriptions, aiming to improve eIDAS2 and NIS2 legal clarity and observance. ArchiMate is used as the modelling language since it provides a structured enterprise architecture framework that visually represents legal requisites. The final architecture descriptions include multiple documented views offering insights for entities seeking to understand and comply with these two legislative instruments, along with practical examples demonstrating their real-world application. Furthermore, work with Artificial Intelligence is explored for validation and documentation of these views within ArchiMate.

## KEYWORDS

Legal Compliance • Conceptual Modelling • ArchiMate • Architecture Description • eIDAS2 Regulation • NIS2 Directive

## 1 Introduction

This work focuses on two key European legal frameworks: the Electronic Identification, Authentication and Trust Services regulation (eIDAS2) and the Network and Information Systems directive (NIS2). Both play a central role in digital trust and cybersecurity across the European Union (EU). By applying conceptual modelling (in this case with ArchiMate), this work aims to create formal architecture descriptions (ADs) that align legal requirements with Enterprise Architecture (EA) practice.

### 1.1 Problem and Motivation

eIDAS2 and NIS2 are key to the digital strategy within the EU. For eIDAS2, a complete structured model is absent in its ecosystem [1], leading to misunderstandings and compliance challenges. As the regulation pushes toward decentralized identity architectures, the need for well-defined models becomes increasingly urgent [2]. In parallel, NIS2 expands to include more stringent requirements, compelling organizations to manage complex cybersecurity needs. This calls for detailed architectural representations to ensure the comprehensive implementation of cybersecurity measures [3]. Moreover, research mentions a notable gap on

structured architectural models for NIS2 implementation, particularly from an EA perspective [4].

Optimistically, previous initiatives already show the value of EA modelling, but no equivalent effort exists for eIDAS2 or NIS2. This creates an opportunity to pioneer structured models (in this case, based on ArchiMate) that ultimately facilitate legal compliance. Nonetheless, the motivation of this work is not limited to creating these two legal ADs, it also aims to demonstrate how can they be applied in real-world cases and how Artificial Intelligence (AI) can support their refinement.

### 1.2 Objective

The main objective of this work was to develop two ArchiMate ADs: one for eIDAS2, and one for NIS2. The goal was to create a structured conceptual model that clearly represents eIDAS2 and NIS2, facilitating their understanding and compliance. Ultimately, the final ADs must capture legal requirements in structured conceptual models, enabling: (1) legal coverage of NIS2 and eIDAS2, ensuring clear understanding of the requirements, (2) traceability, especially between legal clauses and architecture elements, and (3) extraction of independent and documented views.

To support this, practical scenarios implemented in viewpoints and views as extensions to the initial ADs demonstrate how the models can be applied in real-world contexts. Exploratory validation and documentation with emerging AI tools were also tested to assess the potential of "AI-assisted" verification and refinement of large-scale legal models, thereby addressing a well-known limitation of conceptual modelling and EA approaches to complex scenarios.

## 3 Related work

Models that enrich the understanding of legal compliance are lacking. This area is intricate due to its reliance on constantly evolving legislation and the need to balance legal requirements with business goals. The complexity and importance of eIDAS2 and NIS2 call for frameworks that simplify their interpretation.

Despite the absence of complete models representing these legal frameworks, work has been done in related fields to address legal compliance through conceptual modelling. For example, frameworks such as TOGAF have been applied to align EA with legal compliance. A case study demonstrated how the application of TOGAF can facilitate the implementation of the GDPR in small and medium-sized enterprises (SMEs) [5]. Additionally, research introduces an EA viewpoint to assist organizations in implementing the NIST CSF by mapping its seven-step process onto the ADM.

This combined process enables organizations to adopt the NIST CSF within a single architecture development cycle [6].

Below, the state of the art will be further conferred by presenting existing research relevant to this project.

### 3.1 Modelling Identity with ArchiMate

ArchiMate itself provides a structured approach to modelling identity, supported by studies exploring how digital identity can be effectively represented within the language. It involves representing individuals (Actors) and their associated digital identities (Accounts) within an EA framework. In this context, research investigates example frameworks for creating ArchiMate architecture models tailored to digital identity systems. For example, it explores how to integrate Identity and Access Management (IAM) into NIS2 by “mapping NIS2 directive stages” [7] with ArchiMate, providing a structured approach to visualizing compliance.

ArchiMate identity management-related research also includes: (1) a proposed identity management service model framework (IMSMF) that consists of “a meta-model and several models for various protocols and implementations” [8]. By employing ArchiMate, the authors provide a structured approach to represent IAM services, enabling a clear visualization of complex IAM structures and promoting better understanding among stakeholders involved in identity management processes; (2) a proposed universal federated identity management service model (FIMSM), applied to “describe various federated identity management scenarios in a generic service-oriented way” [9], using ArchiMate to represent key elements of their proposed service model. It emphasizes the need for a unified conceptualization of FIM services to deal with complexities in cross-organizational identity management.

### 3.2 Modelling eIDAS2 and NIS2

Representations of eIDAS2 and NIS2 exist, but these tend to be disorganized, difficult to follow, or decentralized, often focusing only on specific parts of the legislation. For instance, the European Digital Identity ARF summarizes the understanding, from eIDAS Expert Group, of the EUDI Wallet concept within a dedicated GitHub repository, which provides detailed architectural models and processes of the EUDI wallet [10]. Another example focuses on a specific area of implementation, such as eID for universities, which applies the eIDAS framework in a university context by building two cross-border eIDAS-enabled academic services based on the eIDAS infrastructure: “the eRegistration in the Erasmus student exchange program and the Login facility with national eIDs on the university portal” [11].

For NIS2, modelling approaches, such as an ontological approach to compliance verification, have been explored to achieve two fundamental objectives: “domain modelling and resource interrogation” [12]. This approach aims to be “helpful for any organization that is going through the hard task of compliance verification.” However, these approaches do not capture the broader legal landscape or the diverse stakeholder perspectives. Additionally, there is an analysis of how NIS2 can be translated into user-centered requirements

to guide real-world security evaluations, presenting a structured, persona-based approach to interpreting NIS2 and facilitating compliance through clear, stakeholder-focused requirements rather than abstract legal clauses [13]. This relates to this work, as it supports the use of structured, stakeholder-driven modelling in ArchiMate to translate NIS2 legal requirements into clear architecture views.

As with eIDAS2, the full scope of NIS2 is often not modelled, and ADs remain limited. Even outside the EU, in the United States, NIST also offers some digital identity models that reflect technologies and architectures already available in the market, such as an “Identity Proofing and Enrollment” model [14]. Reference models often address specific areas such as digital identity, digital wallets, or identity management, which align with legal frameworks but do not fully cover them.

## 4 Method

Creating an AD that models complex legislation while addressing diverse legal concerns presents a significant challenge. The guiding question is: How can we simplify and group legal clauses in a way that keeps all important information while making the AD easy to understand? To answer this, a methodical and structured approach, presented in this section, was followed. This involved breaking down each piece of legislation into smaller parts, organizing them in a clear format, and then translating those parts into ArchiMate views. Nonetheless, all the deliverables discussed and presented from this section onwards are organized within a dedicated GitHub repository [15], which provides a centralized and structured way to manage the project artifacts.

### 4.1 Excel Traceability Matrix

The process began with the creation of a traceability matrix (in Excel), which provides a structured overview of how the legal text from both NIS2 and eIDAS2 can be broken down into identifiable units. Each unit (row) in the matrix represents a specific legal clause, clearly identifying the chapter, article, paragraph, points, and even sub-points (e.g., 1.1.a or 1.2.e). A sample can be seen in **Table 1**.

These units are linked to the ArchiMate AD through a Numbering System, a consistent reference system used throughout the AD, while also serving as a mapper for navigation. Additionally, color coding is used to distinguish between different chapters or types of content, improving visual clarity.

| Chapter    | Article    | ID1 | ID2 | ID3 | ID4 | ID5 | ID6 | ID7 | Text                                                                                                                               |
|------------|------------|-----|-----|-----|-----|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------|
| CHAPTER VI |            | 6   |     |     |     |     |     |     | CHAPTER VI - INFORMATION SHARING                                                                                                   |
| CHAPTER VI | Article 29 | 6   | 29  |     |     |     |     |     | Article 29 - Cybersecurity information-sharing arrangements                                                                        |
| CHAPTER VI | Article 29 | 6   | 29  | 1   |     |     |     |     | 1. Member States shall ensure that entities falling within the scope of this Directive and, (...), where such information sharing: |
| CHAPTER VI | Article 29 | 6   | 29  | 1   | 1   |     | a   |     | (a) aims to prevent, detect, respond to or recover from incidents or to mitigate their impact;                                     |
| CHAPTER VI | Article 29 |     |     |     |     |     |     |     | ...                                                                                                                                |
| CHAPTER VI | Article 30 | 6   | 30  |     |     |     |     |     | Article 30 - Voluntary notification of relevant information                                                                        |
| CHAPTER VI | Article 30 |     |     |     |     |     |     |     | ...                                                                                                                                |

**Table 1:** Sample from the traceability matrix of NIS2

## 4.2 ArchiMate Index

Building on the traceability matrix, the next step was to establish an entry point, in ArchiMate, for each AD: the 'Index' view. While the matrix provides an external reference point for navigating the legislation, the 'Index' view enables internal navigation between the chapters and articles of the legislation and their corresponding architectural representations from a single location. Together, the traceability matrix, the 'Index' view, and the views of the ADs form a "traceability carousel", linking the legal text to its "architecture".

In this 'Index' view, each chapter of the legislation is represented as an ArchiMate group, which contains the corresponding ArchiMate views that model each legislation article. While most articles are individually represented, some have been combined into a single view, subdivided across multiple views, or integrated into other related views. All these variations are easy to understand through the Numbering System found in the documentation for each view element, which contains an identifier linked to the corresponding legal clause. Nevertheless, the view element was used since it is clickable, which permits navigation, allowing for efficient exploration of the AD from a centralized point. The NIS2 view is presented in **Figure 1**.

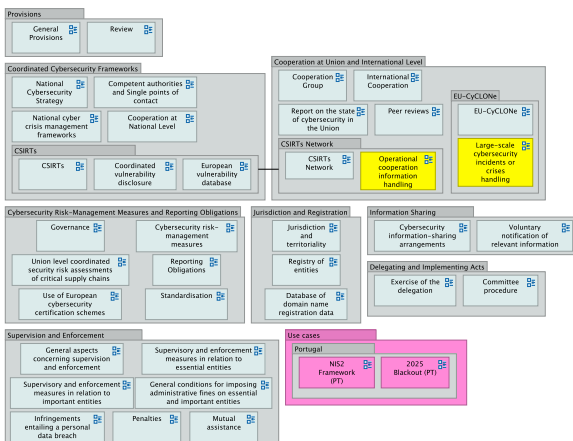


Figure 1: ArchiMate NIS2 Index view

## 4.3 Numbering System

The numbering system, mentioned earlier, is the key traceability feature, with the Excel matrix serving as its foundation. It provides a simple ID mapping between the legislation clauses listed in the matrix and the elements in the AD. Each ID corresponds to a specific "clause level" in the legislation (e.g., chapter, section, article, paragraph, subparagraph, point, sub-point, etc.). Combining both ADs, there are more than one thousand numbered elements. In the traceability matrix, each clause of the legislation is assigned a unique hierarchical ID reflecting its full structural depth. For example, "Article 2, (2), (a), (i)" is represented as "1.2.2.1.1.a.i," capturing all intermediate levels, including nested lists, to ensure precise mapping and avoid ambiguity. This layered format uniquely identifies every clause, no matter how deeply nested.

## 4.4 Building the ArchiMate ADs

The process culminates in the development of the ADs for both NIS2 and eIDAS2. It followed a top-down approach, progressing from a high-level legislative perspective to detailed architectural modelling. For each legislation, a chapter-by-chapter and article-by-article analysis was done, critically evaluating the legal text to identify which elements were relevant and should be included in the architecture. This analysis included the selection of key actors, roles, processes, and many other elements, with a strong focus on the Business layer of ArchiMate. Each article was strictly translated into architecture by adhering to core principles while remaining as close as possible to the legal clauses.

### 4.4.1 Architecture Principles

Given the complexity and granularity of both eIDAS2 and NIS2, careful modelling was required to ensure that the resulting architecture remained clear, interpretable, and faithfully aligned with the legal texts. In the end, each view illustrates a model that facilitates understanding of legal compliance for the stakeholders, and within its design, it adheres to requirements such as the ArchiMate best practices, legal norms, and the ISO/IEC/IEEE 42010 standard. The following architecture principles guided the development of this AD:

- **Abstraction:** Views are at the right level of legal detail, avoiding unnecessary complexity and focusing on what adds value to the understanding of the legal clauses. This is a must since it allows complex legislation to be represented at different levels of detail, helping stakeholders focus on relevant aspects without being overwhelmed by unnecessary complexity;

- **Reuse:** Once legal elements, style types, and architecture patterns were defined and modeled, they were consistently used in all applicable views. For instance: legal elements like data objects were reused where relevant; style types such as labelling, grouping similar cases, arranging relations, and coloring were applied uniformly; architecture patterns like process-driven and event-driven, that captured legal text with complex behavior, allowed for the usage of their corresponding modelled behavioral design across views. Also, anti-patterns, such as generic association relations and usage of invisible properties, were minimized. This ensures coherence and clarity throughout the model, while reducing duplication of effort;

- **Traceability:** As mentioned before, the Numbering System between the matrix and the AD allows tracking back from elements to the legal clauses, or vice versa. Traceability in legal EA modelling is important because it ensures that every architectural element can be directly linked to its source requirement or legal clause, enabling verification, compliance, and easier maintenance of the architecture;

- **Modularity:** The AD generally models one view per article (legal unit), creating separate views if the content is complementary, and organizing elements using Grouping elements. Views themselves are independent (e.g., the ability of each view to represent a specific architectural area, such as the EUDI Wallet, in isolation). This approach enables

stakeholders to understand each view without the need to consult the entire AD. Additionally, the ArchiMate architecture repository is carefully structured into folders for clarity and ease of navigation.

#### 4.4.2 Color table

During the development of views, a color table (**Table 1**) was followed to guarantee visual consistency and to help distinguish between different types of elements within each view.

| Color          | Representation                   |
|----------------|----------------------------------|
| Blue (aqua)    | EU entity or element             |
| Green (fern)   | Portuguese element               |
| Orange (mocha) | Spanish element                  |
| Magenta        | Other relevant entity or element |
| Green          | State or Positive element        |
| Yellow         | State or Complementary element   |
| Orange         | State                            |
| Red            | State or Negative element        |

**Table 2:** Color table used throughout the modelling

Colors in ArchiMate have no formal semantics, leaving their use flexible for enterprise architects to highlight specific elements or emphasize certain aspects as needed. That said, they were used only when they were considered relevant to distinguish something.

#### 4.4.3 Documentation

Each view is thoroughly documented, with its elements referencing official legislative sources, such as the EU Journal (EUR-Lex), thereby ensuring legal accuracy. It complements the numbering system, which already provides basic information about each element, by including additional details or context when needed.

This helps reduce the need to frequently refer back to the matrix or risk missing important nuances, especially since views are independent, and clicking on a single element may require more context than what the article or legal clause alone can provide.

## 5 Results

As a result of the building process, producing a first consolidated version (v1), the ADs provide comprehensive, coherent, and navigable representations of both eIDAS2 and NIS2. However, it is important to acknowledge the inherent difficulty of modelling large and complex legal ADs such as these. With more than a hundred views and over a thousand numbered elements representing different legal clauses, producing perfectly detailed ADs manually is not feasible. In this section, views from both ADs are presented. Views were extracted from each AD to represent every major aspect of both NIS2 and eIDAS2.

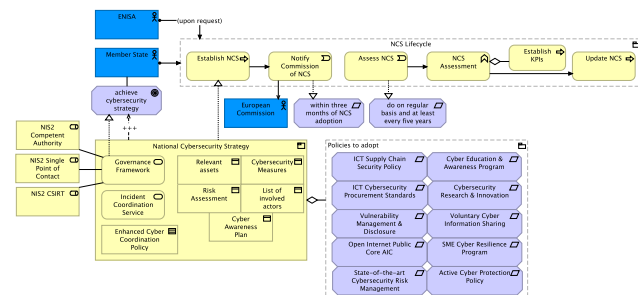
In this way, it is possible to complement the initial discussions, where both legal frameworks were introduced within the full document, by directly presenting their architectures together with some relevant considerations that strictly follow the latest ArchiMate 3.2 specification.

## 5.1 Architecture Description of NIS2

NIS2 AD models the cybersecurity regulatory framework introduced by the directive, focusing on its application to essential and important entities, national CSIRTs, supervisory authorities, and other European-level cooperation institutions. Following two subsections of NIS2 introduced in the full document, a view for both can be extracted from the AD.

### 5.1.1 National Cybersecurity Strategy

This subsection introduces the view (**Figure 2**) representing the NCS as defined by NIS2 in Article 7 [3]. NIS2 requires MSs to establish their National Cybersecurity Strategy (NCS), notify the European Commission, and undergo regular assessments, made by ENISA (upon request), to ensure effectiveness. ENISA supports this assessment by establishing KPIs and updating the NCS as needed. The strategy includes a governance framework, risk assessments, cybersecurity measures, and awareness plans, complemented by policies like ICT supply chain security, cybersecurity education, and SME resilience programs. This cohesive approach ensures a robust cybersecurity posture across the EU.



**Figure 2:** View of the national cybersecurity strategy

#### ArchiMate architectural comments:

The NCS can be modeled as a business product, as it encapsulates a coherent collection of services and objects, accompanied by a contract. ArchiMate enables a logical representation of this structure. The relevant policies are modeled as requirement elements within the motivation layer, as they express statements of need that justify and guide the development of the NCS. Additionally, the lifecycle of the NCS is represented through business processes, illustrating how its strategy evolves over time while maintaining alignment with plans from the EU. Also relevant to say that grouping in ArchiMate is always a good option when aggregating concepts that belong together based on some common characteristic, which is why it is used a lot on both ADs.

### 5.1.2 Incident Reporting and Sector Coverage

This subsection illustrates the workflow (**Figure 3**) of reporting obligations defined in NIS2, in Article 23 [3]. Incident notification and reporting require entities to promptly notify CSIRTs of significant incidents that cause severe disruption or harm. When an incident occurs, entities

```

graph TD
    Incident[Incident] --> SignificantIncident[Significant Incident]
    Incident --> Voluntary[Addition on 'voluntary notification of relevant informant' view]
    SignificantIncident --> NotifyCSRT[Notify CSRT]
    NotifyCSRT --> EarlyWarning[Early warning of potential malicious or cross-border incident]
    EarlyWarning --> Answer[Answer to the notifying entity]
    Answer --> NSQCSRT[NSQ CSRT]
    NSQCSRT --> Public[public, transparency principle]
    NSQCSRT --> ReportRequest[Report request]
    ReportRequest --> ReportIncident[Report incident/impact information]
    ReportIncident --> IntermediateReport[Intermediate Report]
    IntermediateReport --> FinalReport[Final Report]
    FinalReport --> NotLater[not later than one month]
    SignificantIncident --> NotifyRecipients[Notify recipients of service]
    NotifyRecipients --> Ankyr[Ankyr]
    Ankyr --> NSQCSRT
    Ankyr --> ReportRequest
    Ankyr --> ReportIncident
    Ankyr --> TADA[TADA]
    TADA --> InformFindings[Inform findings on notifications]
    InformFindings --> CooperationGroup[Cooperation Group]
    InformFindings --> NSQNetwork[NSQ CSRT Network]
    InformFindings --> SummaryReport[Summary Report]
    InformFindings --> SinglePoint[NSQ Single Point of Contact]
    SinglePoint --> StatusSummary[Status Summary Report]
    StatusSummary --> EveryThreeMonths[every three months]
    TADA --> NotifyVulner[Notify vulner]
    NotifyVulner --> ImpactMTU[Impact > 1 MTU]
    ImpactMTU --> NotifyCPS[Notify CPS or cross-sectoral impact of the incident]
    NotifyCPS --> DetermineBorder[Determine cross-border or cross-sectoral impact of the incident]
    DetermineBorder --> Within72[within 72 hours]
    Within72 --> IncidentNotification[Incident notification (update)]
    IncidentNotification --> Public
    IncidentNotification --> ReportRequest
    IncidentNotification --> ReportIncident
    IncidentNotification --> TADA
    TADA --> Within24[within 24 hours]
    Within24 --> EarlyWarning
    TADA --> WhereApplicable[where applicable, MS competent authority]
    WhereApplicable --> NSQCSRT
    WhereApplicable --> ReportRequest
    WhereApplicable --> ReportIncident
    WhereApplicable --> TADA
    TADA --> EverySixMonths[every six months]
  
```

**ArchiMate architectural comments:**

Moreover, to represent conditional triggers, when needed, these relationships are named with conditions, thereby facilitating the reading of this complex ArchiMate behavioral view.

The eIDAS2 AD addresses the expanded legal and technical scope introduced by the revision of the original eIDAS. It covers the legal framework for digital identity, including its means, methods, and mechanisms for secure identification, authentication, and trust services across the EU. Following the two subsections of eIDAS2 introduced in the full document, a view for both can be extracted from the AD.

This subsection introduces the view (**Figure 4**) that represents eID and cross-border interoperability under Article 11a of the eIDAS2 regulation [16]. Cross-border identity matching is a clear example of cross-border interoperability through electronic identification. This view shows the interaction between the user and the relying party (RP), in the context of access to a cross-border service. Users use their notified electronic identification means or EUDI Wallet instance to access a cross-border service. The MS acts as a RP, ensuring cross-border identity verification, with unequivocal identity matching. This process involves validating the PID of the user, providing access to the cross-border service while implementing data protection measures, such as GDPR. The MS where the eID or Wallet originates is the entity that generates and provides the PID to the RP.

```

graph LR
    User[User] -- "(using)" --> EID[Electronic Identification Mean]
    User -- "(using)" --> EUDI[EUDI Wallet Instance]
    User --> MS[MS Acts as relying party in cross-border service]
    User --> CB[Cross-border service]
    MS --> RP[Relying Party]
    RP --> MS
    RP --> MS_M[Member State]
    MS_M --> RP
    CB --> CBI[Cross-border identity matching]
    CBI --> UIM[unequivocal identity matching]
    UIM --> AP[Access provision to cross-border service]
    CB --> CBI_V[Cross-border identity verification]
    CBI_V --> IDP[Implement data protection measures]
    IDP --> PID[PID]
    IDP -.-> CBI_V
    PID -.-> IDP
  
```

The diagram illustrates the cross-border identity matching process flow. It involves a User, MS (Member State) acting as a relying party, a Relying Party, and a Cross-border service. The process includes steps for using electronic identification means and EUDI wallet instances, performing cross-border identity matching and verification, and implementing data protection measures. The flow is as follows: User uses EID and EUDI to access the Cross-border service. The Cross-border service performs identity matching and verification. The Relying Party interacts with the MS and the Cross-border service. The Cross-border service implements data protection measures and provides access to the cross-border service.

### ArchiMate architectural comments:

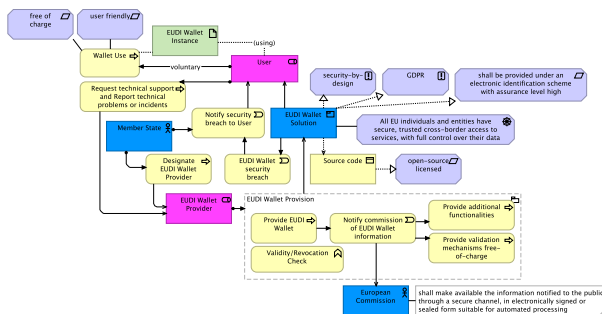
It is also important to highlight the unequivocal identity matching principle in cross-border identity verification, as it expresses a general property that governs and applies to this process.



### 5.2.2 Digital Wallet

This subsection explores views related to the EUDI Wallet, the central innovation of eIDAS2, addressing its Articles 5a and 5c [16]. For instance, **Figure 5**, shows that the provision and use of the EUDI Wallet are voluntary for users, who can request technical support and report incidents to the MS. MSs notify users of security breaches and manage the security of the EUDI Wallet. The wallet solution adheres to GDPR and security-by-design principles, ensuring secure, trusted, and seamless cross-border access to services while maintaining user control over data, while serving the user, each one within their wallet instance. The source code of the wallet is open-source licensed.

MSs designate an EUDI Wallet provider that comes up with the EUDI Wallet, conduct validity and revocation checks, notify the European Commission of wallet information, and provide validation mechanisms free of charge and additional functionalities. The Commission shall make available the information notified to the public through a secure channel, in electronically signed or sealed form suitable for automated processing.



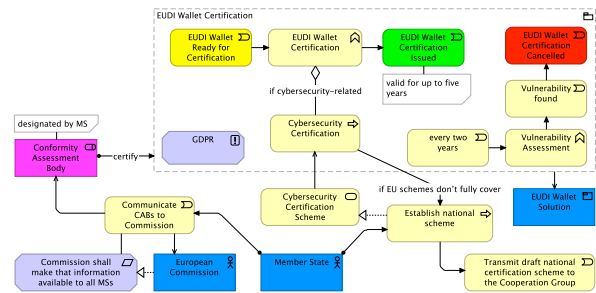
**Figure 5: View of the EUDI Wallet**

### ArchiMate architectural comments:

This is an example of a complex and extensive translation of a legal article into ArchiMate. While it may lack some readability, it remains concise and effectively shows the major business behaviors associated with the EUDI Wallet Solution, as well as its interactions with key actors, the user, the European Commission, and MSs. It covers scenarios such as wallet usage, provision, support, and provider designation. Although not immediately clear at first glance, the model maintains overall coherence and consolidation.

eIDAS2 also introduces a comprehensive framework for the certification and use of the EUDI Wallet, as shown in **Figure 6** (Article 5c). The certification process begins with the EUDI Wallet being ready for certification, overseen by a CAB designated by MSs. This certification must comply with GDPR standards. If the certification is cybersecurity-related, it must undergo a cybersecurity certification, served by an EU cybersecurity certification scheme. If this EU scheme doesn't fully cover the certification needs, the national MS must establish a national scheme and transmit it to the Cooperation Group. Once the certification is issued, the certification is valid for up to five years, subject to

vulnerability assessments every two years. If vulnerabilities are found, the certification can be cancelled.



**Figure 6:** View of the certification of EUDI Wallets

**ArchiMate architectural comments:**

In ArchiMate, colors are used precisely for situations like the figure above, where they help the reader quickly associate elements with the stages of EUDI Wallet certification - Ready, Issued, and Cancelled.

## 6 Demonstration

With both ADs now presented, it becomes clear that their primary purpose is to ease, structure, and clarify the business and motivational understanding of both eIDAS2 and NIS2, while also guiding enterprise architects in legal modelling. However, their utility extends further.

This section will introduce a demonstration of the work, showcasing the practicality of the results obtained from the method. That said, views describing practical scenarios for hypothetical stakeholders were developed (included in the v1 of the results) to demonstrate the applicability of these views in real-world cases, addressing concrete concerns derived from legislative requirements and enabling stakeholders to better understand, implement, and comply with NIS2 and eIDAS2 obligations.

These views are built directly within the ADs, reusing existing elements and documentation, representing extensions to the architecture of both legal frameworks, while following the previous Architecture Principles. Below, only two scenarios from the full document will be shown.

## 6.1 Practical Scenario

The definition and presentation of practical scenarios in this section strictly follow the ISO/IEC/IEEE 42010 standard, regarding the organization of concepts. These scenarios are purely illustrative and demonstrative. They were not submitted to any of the stakeholders mentioned, although they were developed with the awareness that such cases could be relevant and of potential interest to them.

The process began with identifying stakeholders, a critical step in ensuring the relevance and success of these practical scenarios. Then, efforts were made to address their possible key interests and specific concerns. Afterwards, these were grouped to help define relevant viewpoints through classification. Subsequently, the legal environment was modeled by incorporating elements from the respective legislation AD. Each view was carefully designed to address all identified concerns tied to its viewpoint. Once the views

of practical scenarios were developed, the refinement process started, incorporating necessary changes and addressing any newly identified concerns. This approach ensured that the final views for practical scenarios were fully aligned with stakeholder expectations and requirements.

## 6.2 Cybersecurity Perspective of 2025

### Blackout

The 2025 blackout event scenario, in **Figure 7** and **Figure 8**, offers a practical context for examining how the Portuguese compliance with NIS can strengthen national resilience and compliance with EU cybersecurity standards since the directive has been approved by the Portuguese Parliament but is not yet fully in force, awaiting official publication in the *Diário da República*. Therefore, this scenario represents an example of how the blackout events can be aligned with NIS2 and be useful for an entity like CNCS.

Portugal experienced a historic blackout that began on April 28 at 11:30 AM, causing disruption across transport, telecommunications, and healthcare. The *Centro Nacional de Cibersegurança* (CNCS) quickly ruled out a cyberattack and coordinated with Spain, France, the European Commission, and ENTSO-E to investigate. On April 29, the civil group Citizens for Cybersecurity (CpC) reported the alleged sale of access to energy infrastructure on the dark web, prompting a formal complaint and requests for telecom accountability. By April 30, CNCS reiterated the absence of malicious activity, while INCIBE, from Spain, began a technical investigation with Portuguese authorities. On May 1, Anacom issued recommendations to improve emergency communications, and the government ordered an urgent review of the SIRESP system.

On May 9, the report from ENTSO-E confirmed that frequency instabilities in Spain triggered the blackout, with no evidence of cyber compromise. In parallel, Portugal advanced the transposition of the NIS2 Directive, reinforcing obligations for critical infrastructure, and on June 25, the government unveiled new energy security measures, including expanded "black start" capabilities and battery installations in key facilities.

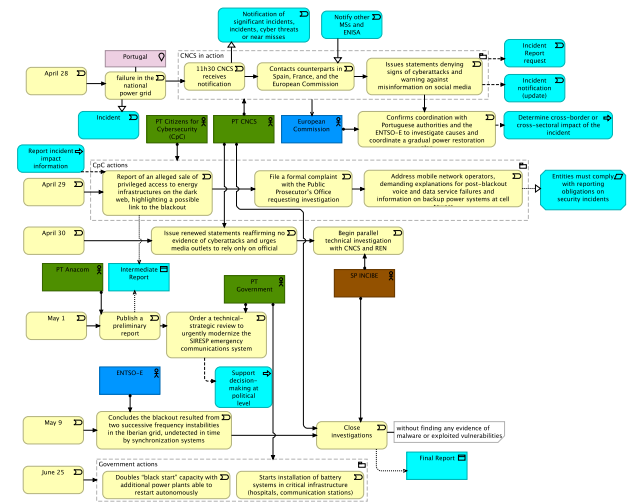
**Major Stakeholder:** CNCS

**Viewpoint:**

| Cybersecurity Perspective of 2025 Blackout Viewpoint |                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stakeholders                                         | 1. CNCS                                                                                                                                                                                                                                                                                         |
| Concerns                                             | 1. Overall response of Portuguese entities to the Blackout<br>2. Core behavior and timeline of events<br>3. Systemic risk of large-scale power outage and its cascading effects on critical sectors.<br>4. Incident response coordination and communication between national and EU authorities |
| Purpose                                              | Designing                                                                                                                                                                                                                                                                                       |
| Abstraction Level                                    | Coherence                                                                                                                                                                                                                                                                                       |
| Layer                                                | Business                                                                                                                                                                                                                                                                                        |
| Aspects                                              | Active structure, Behavior, Passive structure                                                                                                                                                                                                                                                   |

**Figure 7:** Viewpoint of the cybersecurity perspective of the 2025 blackout

**View:**



**Figure 8:** View of the cybersecurity perspective of the 2025 blackout

### ArchiMate architectural comments:

The Blackout behavior represents a huge sequence of events in different days, so different flows were made for each day. This separation supports a more accurate and simple representation of the operational timeline. Elements related to NIS2 are highlighted in **cyan** to illustrate clearly NIS2 elements from the original AD as a guiding framework for these cases. All non-cyan elements represent business events, as they are not specific processes described in the directive.

Moreover, implementing this complex case shows how ArchiMate can model multi-day, multi-actor operational events, capturing both the temporal sequence and interactions between national authorities, international bodies, and civil groups, while maintaining clarity in the representation of cascading effects and responses, although the layout lacks some readability.

## 6.3 eIDAS2 Framework in Portugal

This scenario represented in **Figure 9** and **Figure 10** presents an architectural overview of the eIDAS2 framework in Portugal, similar to what was done for NIS2. As depicted in the view, AMA is the eIDAS2 National Authority and SPC, reason for being the major stakeholder of this view. AMA coordinates with GNS, which has the eIDAS2 supervisory role. It is also responsible for both the supervision and development of the Portuguese Digital Identity Wallet and the Portuguese electronic identification scheme, CMD.

Relevant relationships to the Portuguese eIDAS2 scenario: The TSP is supervised under the GNS and certified by a CAB, which is accredited and monitored by the IPQ, the National Accreditation Body (NAB) of Portugal. GNS create and send a report summary of trust services notifications, from TSPs, to both ENISA and CNPD. CABs perform compliance assessments to the PT Wallet. Furthermore, AMA is connected to broader European oversight through the European Commission and other MSs, aligning with eIDAS2 interoperability.

## Major Stakeholder: CNCS

### Viewpoint:

| eIDAS2 Framework Viewpoint |                                                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stakeholders               | 1. AMA                                                                                                                                                        |
| Concerns                   | 1. Compliance with the eIDAS2 regulation at the national level<br>2. eIDAS2 Portugal environmental context<br>3. Major Portuguese actors and roles for eIDAS2 |
| Purpose                    | Informing                                                                                                                                                     |
| Abstraction Level          | Coherence                                                                                                                                                     |
| Layer                      | Business                                                                                                                                                      |
| Aspects                    | Active structure, Behavior, Passive structure                                                                                                                 |

Figure 9: View of the eIDAS2 framework in Portugal

### View:

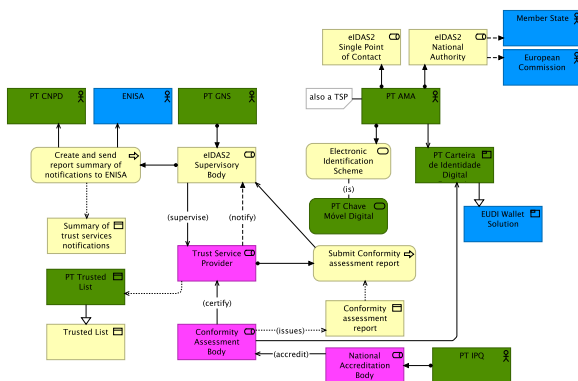


Figure 10: Viewpoint of the eIDAS2 framework in Portugal

### ArchiMate architectural comments:

The flow relation from AMA to MS and the European Commission indicates an information flow or interaction (not control), showing data/report flow.

Also, although cases like the accreditation of the CAB by the NAB could have been modeled as business processes, a more generic representation was used to maintain a high-level perspective in this view.

## 7 Validation and Documentation

Exploratory work with AI was conducted to validate and document ArchiMate ADs. Two preliminary and complementing approaches were explored: a unit-based review with ChatGPT and an AI-enhanced JArch integration with LLaMA and Gemini. These experiments did not aim to replace formal validation but to test experimental methods that point toward possible future directions of EA.

### 7.1 Unit-Based Review with ChatGPT

First, an exploration with ChatGPT 5, from OpenAI, was conducted through unit-based reviews. In this approach, small “units” of the AD, such as views linked to only one article, were selected and provided to the AI together with their traceability information and context. The goal was to test whether ChatGPT could analyze these units step by step and suggest improvements, rather than attempting to process the full model at once.

To represent this technique in the context of ISO 42010, an extension diagram (Figure 11) was created based on its original Conceptual model of an AD. The illustration in Figure

42 represents how AI is integrated into the ISO 42010 conceptual model to support unit-based validation of the AD. At the top, the diagram shows the process flow: starting from a specific given context (legal text, traceability information, and explanations), the information is sent and processed by ChatGPT 5. This feeds into a review response, which provides the actor with the basis to then carry out the unit validation, checking the quality and consistency of the given architectural view or “unit”, and updating if necessary. Once it is done for every unit, the whole AD is AI-validated.

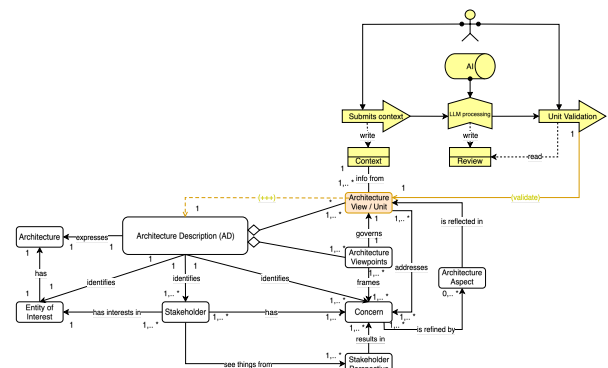


Figure 11: AI Unit-based review extension to ISO 42010 conceptual model of an AD

From this point, unit-based validation with AI is exemplified to show how it was used to refine certain views of the first consolidated version of the AD (v1). Initially, a simple view representing a small legal article was selected. The chosen view, seen in Figure 12, comes directly from the eIDAS2 AD: “EUDI Wallet-Relying Parties” (Numbering 2.1.5b, representing Chapter 2, Section 1, Article 5b).

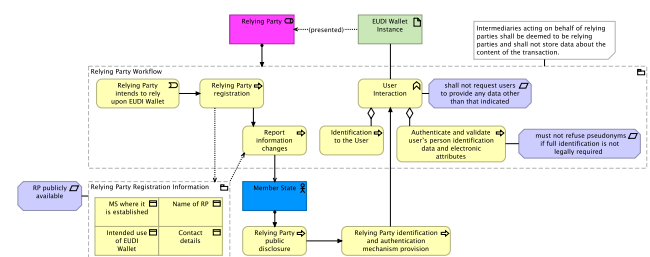


Figure 12: View of EUDI Wallet relying parties (on v1)

The XML of this view was provided to ChatGPT 5, along with the corresponding entries from the traceability matrix, the original legal text, and a contextual explanation of how the numbering system works. Then, it was written what the AI should analyze. The instructions were for the AI to first generate a brief overview paragraph, then produce a small table linking the legal clauses in the matrix to the respective elements. Next, it should assess the validity of the view and whether it strictly follows the legal text.

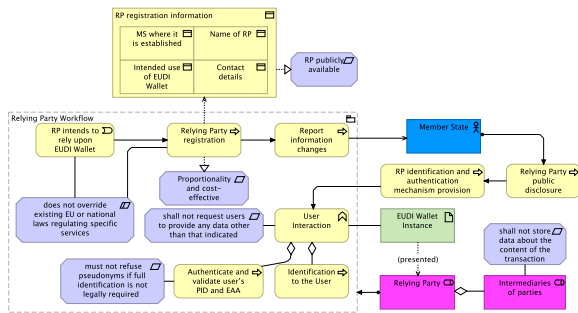
Finally, it was asked to provide ArchiMate-specific comments on what aspects of the view are well-modeled and what should be updated. Figure 13, one of the used prompts.



Annexes: XML and PNG of “EUDI Wallet-Relying Parties” view

As a bonus in this version, some AI considerations were also requested on layout validation. For any enterprise architect, it is important that a view is clear, concise, and well-centralized. The recommendations from AI proved valuable in addressing spatial organization and readability issues, successfully guiding improvements in problematic line crossings, element sizing, and visual grouping. However, the analysis also revealed a critical limitation: the AI struggled to interpret some enterprise architecture

practices, particularly with respect to ArchiMate notation standards. The outcome was the version 2.1.



**Figure 15:** Updated view of EUDI Wallet relying parties (on v2.1)

For the version 2 of this view, ChatGPT was asked to produce documentation to assess the viability and possibilities of adding comments to the views automatically, using the unit approach, based on the legal text and its knowledge of the traceability matrix and numbering system. Some numbering omissions were detected, but minor and not critical. Overall, the documentation provided by ChatGPT was on point - concise and assertive.

Interestingly, in another test run with similar inputs but a more extensive context, ChatGPT succeeded in recognizing more elements and producing documentation with greater detail and accuracy, without losing precision, highlighting the value of iterative and expert use of AI and the importance of maintaining a good traceability system when designing legal ADs. Nonetheless, it must be emphasized that documentation with AI through unit-based review is an excellent approach, but it requires a well-established AD and a traceability system linking views to the legal clauses. Providing views without this system, unconsolidated, or not extremely legal aligned does not produce good results nor giving a bad context or a lot of views.

After these promising results, some of the initial v1 views, along with the validated v2 views, were submitted to this process to refine elements whose documentation was missing or unclear, whether because the legal clause alone was insufficient, or it represented an extra element, ultimately resulting in the v3 of the ADs.

## 7.2 AI-enhanced JArchi

JArchi is a JavaScript extension for the Archi tool enabling AI-assisted ArchiMate modelling. Two approaches were integrated: (1) the “archi-scripts” GitHub library<sup>1</sup>, providing bulk operations, model analysis, AI-assisted generation, documentation, and reporting; (2) GitHub Gists<sup>2</sup> from a developer leveraging the Gemini API, offering automated validation critique, documentation, and assessment of ArchiMate views. Both approaches enhance modelling efficiency and insights, though they have limitations, including a steep learning curve, dependence on configuration and external libraries, variable AI output

quality, and potential misalignments in generated or updated views.

For brevity, full technical details are provided in the complete document. Two scripts were applied to the same version of the previously used view (it’s v2): a Documentation Augmenter using LLaMA (Meta) and a Validation Critique using Gemini (Google).

## 8 Conclusion

Early in the research process, a significant gap was identified: the absence of structured models that fully structure eIDAS2 and NIS2. Building on it, a research question was internally formulated: “How can we simplify and group legal clauses in a way that keeps all important information while making an AD easy to understand?”. Through an in-depth analysis of both legal frameworks, and the ArchiMate language, it was possible to develop a method that allows for the traceability and visualization of legal clauses as architectural elements, simplifying the understanding of complex legal compliance requirements within EA conceptual modelling, filling the research gap.

This work proved effective in systematically transforming the legal requirements of eIDAS2 and NIS2 into ArchiMate elements. For the future, primarily, any improvements to this work - whether by enhancing the methodology, demonstration, or validation approaches and applying them to other legal frameworks, or by refining the current ADs - represent a consolidated step forward.:

Looking ahead, the integration of AI-powered scripts presents a wide range of opportunities for advancing architectural modelling. This evolution suggests a paradigm shift in which enterprise architecture modelers may increasingly rely on intelligent assistants to accelerate model creation, improve reporting, enhance analytical depth, and support systematic validation and documentation. Such an approach opens the way for richer and more adaptive modelling practices in both academic and professional contexts.

Further work should also focus on evolving the “unit” size of the unit-based review approach presented in this study. This would enable more effective comparison of different views across the ADs and their escalation into broader legal contexts, allowing interdependencies between regulations to be better captured. Another key direction is the pursuit of complete legal validation, ensuring that the models not only represent compliance requirements accurately but also withstand rigorous legal interpretation and practical application.

New research could explore alternative methods for improving traceability and the numbering system used in legislative modelling. For instance, experimenting with hierarchical or semantic numbering schemes, dynamic linking between clauses and architectural elements, or integrating more advanced cross-referencing techniques could enhance navigability and automation. Additionally,

<sup>1</sup> ThomasRohde, 2023. “archi-scripts” GitHub repo. <https://github.com/ThomasRohde/archi-scripts#>

<sup>2</sup> Mileham, Steven. 2025. AI support for Archi Models using Gemini and jArchi. General Geekery blog, July 2, 2025. <https://smileham.co.uk/2025/07/02/ai-support-for-archi-models-using-gemini-and-jarchi/>

approaches that systematically compare and reconcile different views across multiple legislative instruments could provide richer insights into interdependencies and overlaps between legal clauses, supporting more robust compliance modelling. Such refinements would further strengthen the practical applicability and accuracy of ADs in legal and enterprise contexts.

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